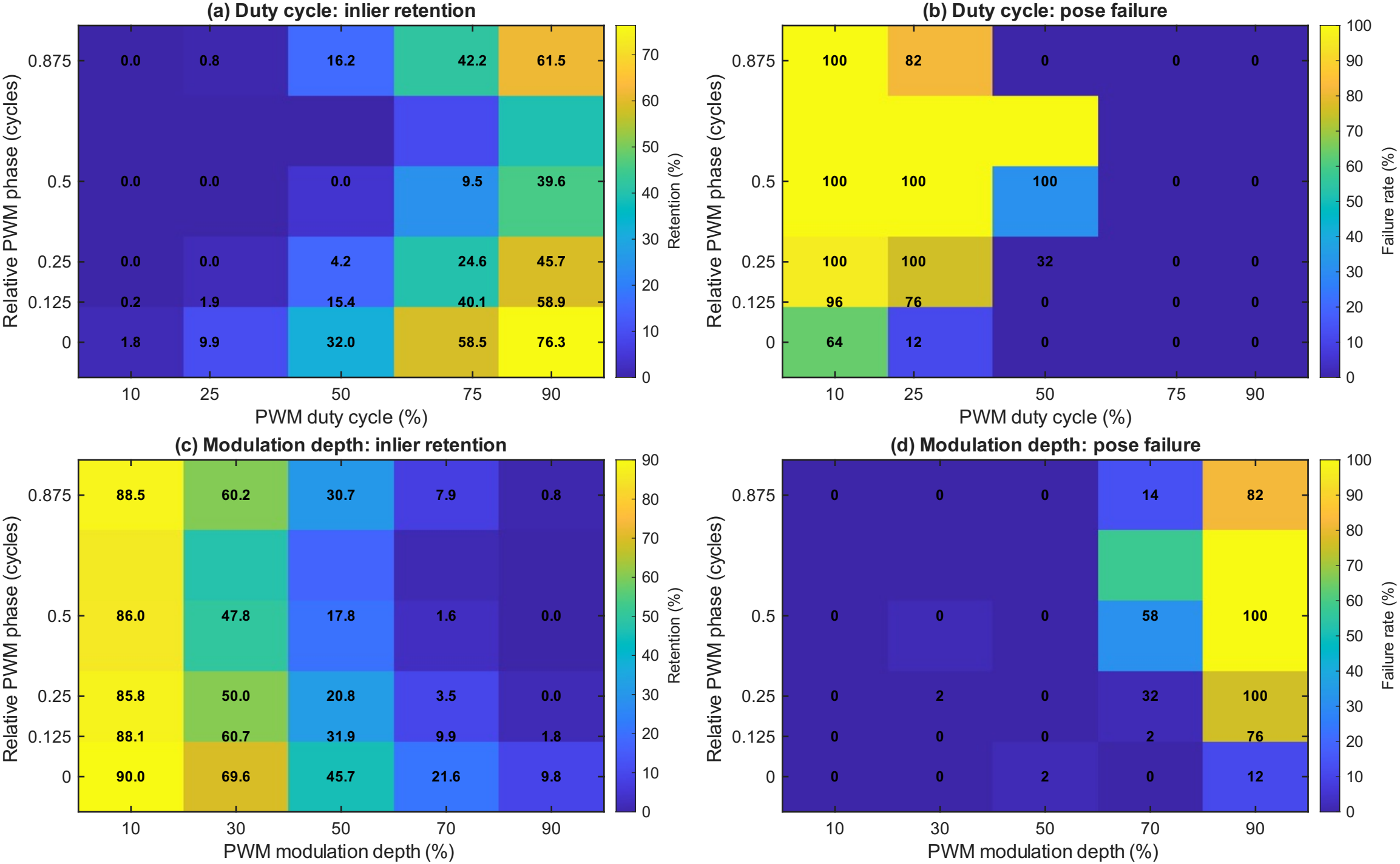

(a) Duty cycle: inlier retention
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
0.0 0.8 16.2 42.2 61.5
0.0 0.0 0.0 9.5 39.6
0.0 0.0 4.2 24.6 45.7
0.2 1.9 15.4 40.1 58.9
1.8 9.9 32.0 58.5 76.3
10 25 50 75 90
PWM duty cycle (%)
Retention (%)
0 10 20 30 40 50 60 70
(b) Duty cycle: pose failure
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
100 82 0 0 0
100 100 100 0 0
100 100 32 0 0
96 76 0 0 0
64 12 0 0 0
10 25 50 75 90
PWM duty cycle (%)
Failure rate (%)
0 10 20 30 40 50 60 70 80 90 100
(c) Modulation depth: inlier retention
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
88.5 60.2 30.7 7.9 0.8
86.0 47.8 17.8 1.6 0.0
85.8 50.0 20.8 3.5 0.0
88.1 60.7 31.9 9.9 1.8
90.0 69.6 45.7 21.6 9.8
10 30 50 70 90
PWM modulation depth (%)
Retention (%)
0 10 20 30 40 50 60 70 80 90
(d) Modulation depth: pose failure
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
0 0 0 14 82
0 0 0 58 100
0 2 0 32 100
0 0 0 2 76
0 0 2 0 12
10 30 50 70 90
PWM modulation depth (%)
Failure rate (%)
0 10 20 30 40 50 60 70 80 90 100